

生产级大模型应用开发平台：LangSmith

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 - LangSmith 是什么
 - LangSmith 账号申请与开发环境搭建
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 - LangSmith 平台功能介绍：Tracing
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 - 实战 LangSmith Tracing
 - 实战 LangSmith Evaluation

LangSmith 是什么？

Get your LLM app from prototype to production

An all-in-one developer platform for every step of the LLM-powered application lifecycle, whether you're building with LangChain or not.



LangSmith 是一个用于构建**生产级** LLM 应用程序的**平台**。通过提供可视化监控和评估 LLM 应用原型的平台，LangSmith 提升了开发者效率，且**支持独立运行**。

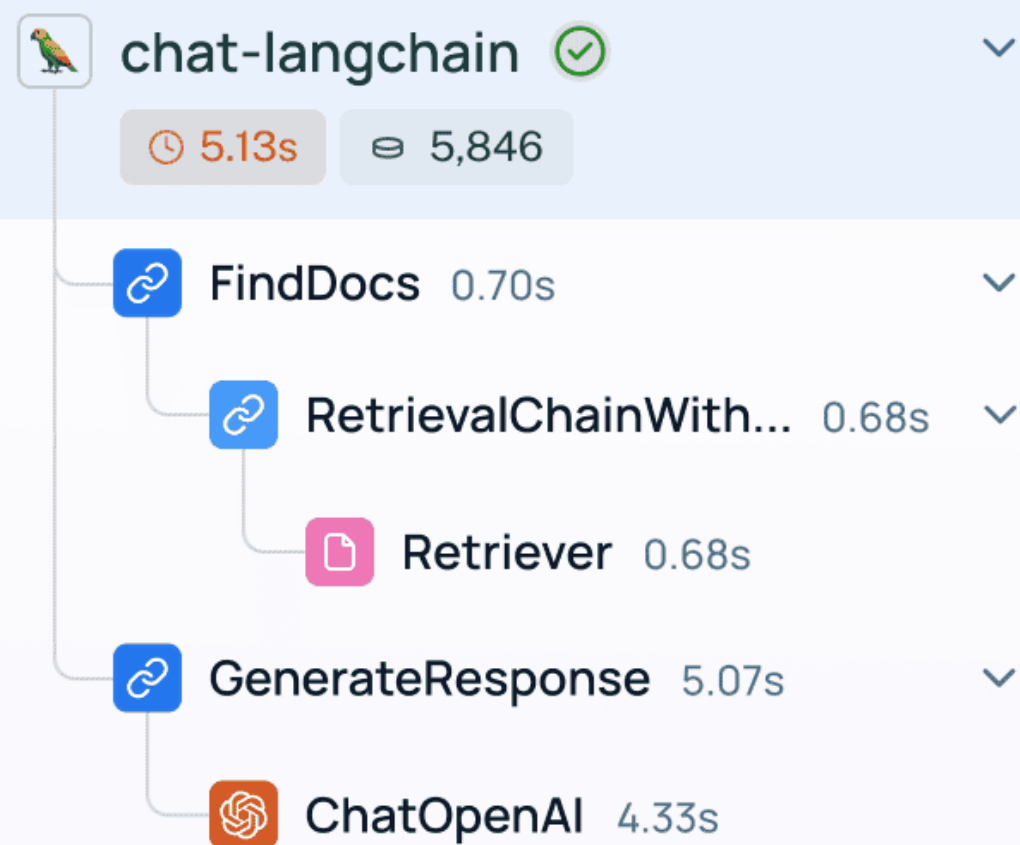
LangSmith 核心功能：

- Tracing: 记录追踪 LLM 生成内容和调用情况；
- Monitoring: 通过 LangSmith 平台可视化 Tracing 记录各项指标；
- Datasets: 为 LLM 应用原型提供各类数据集及其管理功能；
- Evaluation: LLM 应用性能、多样性、准确性等各项评估，需要 Datasets 支持。

LangSmith 注意事项：

- LangSmith 平台：<https://smith.langchain.com/>，需要注册账户和创建 API_KEY。
- LangSmith Client SDK：用于交互 LangSmith 平台的客户端，目前支持 Python 和 TypeScript。
- LangSmith 与 LangChain 类似，还处于高速发展状态，很多功能和 API 在不断完善。

TRACE



chat-langchain

✓ SUCCESS 5.13s 5,846

👤 HUMAN 09:01:00

How do I use a RecursiveUrlLoader to load content from a page?

🧠 AI 09:06:13

To use a RecursiveUrlLoader to load content from a page, follow these steps:

Initialize the RecursiveUrlLoader with the target URL and other optional parameters such as max depth, asynchronous loading, extractor function, and more.

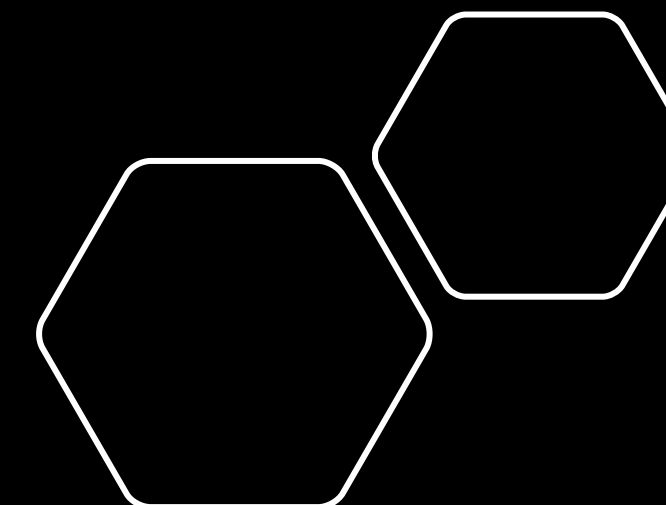
Use the `load()` method to load web pages, which returns a list of documents

Optionally, you can use the `lazy_load()` method for lazy loading or the `load_and_split()` method to load documents and split them into chunks.

📄 Add to Dataset

✍️ Annotate

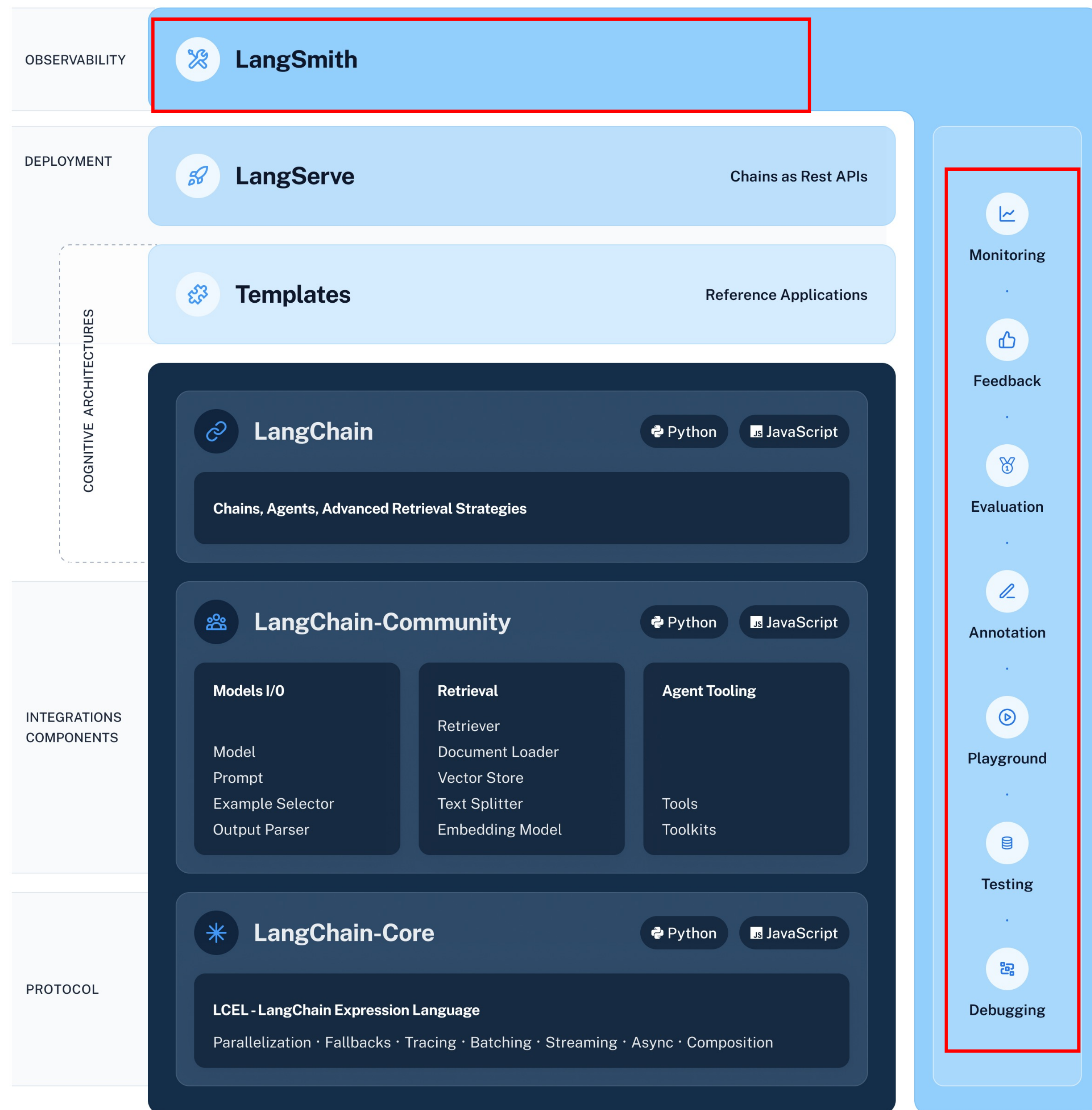
🎮 Playground



LangChain

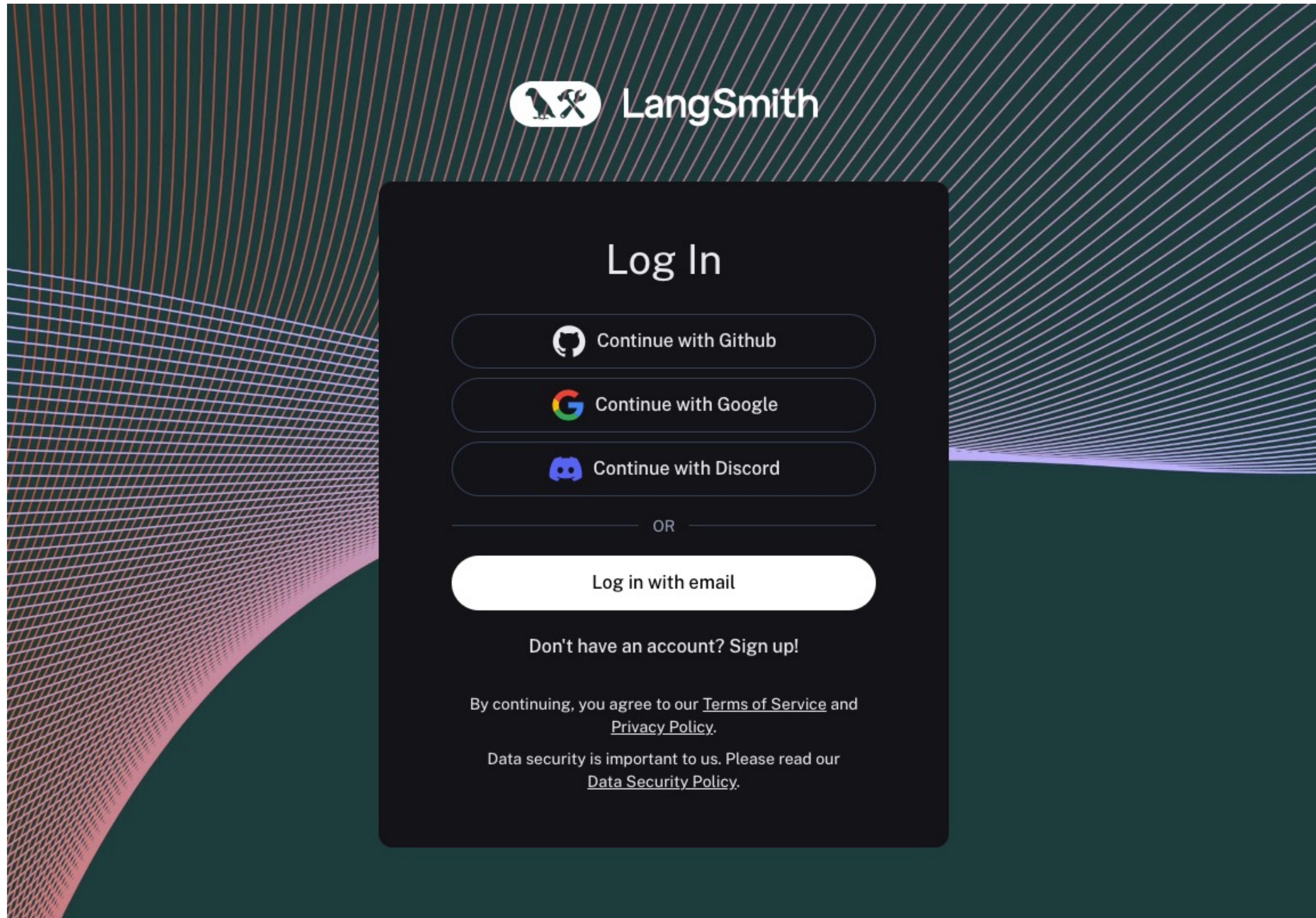
技术生态与社区

- **LangSmith**: 生产级 LLM 应用开发平台
- **LangChain**: LLMs 应用开发框架
- **LangChain-Community**: 第三方集成
- **LangChain-Core**: LCEL 等协议
- **LangChain Templates**: 开箱即用APP示例
- **LangServe**: Chains 生产部署 (REST API)



LangSmith 账号申请与开发环境搭建

LangSmith 平台账号申请



-  Projects 1
-  Annotation Queues 0
-  Deployments
-  Datasets & Testing 1
-  Prompts 0

Projects

 Projects

1 project

[+ New Project](#)

Log runs from your LLM applications

Datasets & Testing

 Datasets & Experiments

1 dataset

[+ New Experiment](#)

Run a prompt/chain over a dataset

 Upload CSV Dataset

Upload a CSV of inputs and outputs

 Export Dataset

Export a dataset in CSV or OpenAI JSONL format

登录 LangSmith

[+ New Annotation Queue](#)

Log runs from your LLM projects

- Projects 1
- Annotation Queues 0
- Deployments
- Datasets & Testing 1
- Prompts 0

- Color Scheme
- Settings
- Documentation
- LangChain Hub

API Keys

API keys

You can only access an API key when you first create it. If you lost one, you will need to create a new one.

Create API Key API docs

- Members
- Models
- Shared URLs
- Secrets
- Feedback Tags
- Rules

Personal Access Tokens are available now! All previous API keys have been converted to Service Keys. You can find more information [here](#)

- Personal
- Service
- All

1 key

Key	Description	Created At	Actions
lsv2_pt_5...8a78	test	5/22/2024, 11:57:19 AM	

创建 API KEY


```
# OpenAI configs
```

```
export OPENAI_API_KEY="sk-7Nnl...xVMC"
```

```
# LangSmith configs
```

```
export LANGCHAIN_TRACING_V2=true
```

```
export LANGCHAIN_API_KEY="l...8a78"
```

LangSmith

环境变量设置

LangSmith 服务协议与计价规则

Startups

Designed for early stage startups building AI applications

Reach out for starter pricing and get shipping today

Reach out

What to expect:

We want all early stage companies to build with LangSmith. LangSmith for Startups offers discounted prices and a generous free, monthly trace allotment, so you can have the right tooling in place as you grow your business.

Developer

Designed for hobbyists who want to start their adventure solo

Free for 1 user

5K traces per month included, pay as you go thereafter ⓘ

Get Started

Key features:

- 1 Developer seat
- Debugging traces
- Dataset collection
- Testing and evaluation
- Prompt management
- Monitoring

Plus

Everything in Developer, plus team features, better rate limits and data retention

\$39/user per month

10K traces per month included, pay as you go thereafter ⓘ

Get Started

Key features:

- All features in Developer tier
- Up to 10 seats
- Hosted LangServe (beta)
- Longer data retention
- Higher rate limits
- Email support

Enterprise

Designed for teams with more security, deployment, and support needs

Custom

Request a demo

Key features:

- All features in Plus tier
- Custom Single Sign On (SSO)
- SLA
- Self-hosted deployment options
- Custom rate limits
- Team trainings
- Shared Slack channel

	Developer	Plus	Enterprise
Features			
Debugging Traces	✓	✓	✓
Dataset Collection	✓	✓	✓
Human Labeling	✓	✓	✓
Testing and Evaluation	✓	✓	✓
Prompt Management	✓	✓	✓
Hosted LangServe	—	In beta	In beta
Monitoring	✓	✓	✓
Team			
Developer Seats	Maximum 1 seat Free	Maximum 10 seats \$39 per seat/month	Custom pricing
Collaborator Seats	—	—	Coming soon!

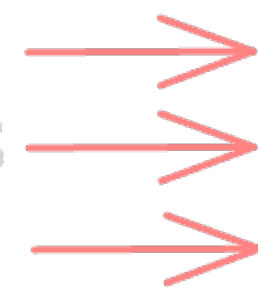
	Developer	Plus	Enterprise
Usage			
Traces	First 5k traces per month for free. \$0.005 per trace thereafter	First 10k traces per month for free. \$0.005 per trace thereafter	Custom
Max Ingested Events / Hour	50k / 250k (with payment method on file)	500k	Custom
Total Trace Size Stored / Hour	500MB / 2.5GB (with payment method on file)	5GB	Custom
Security Controls			
Role Based Access Control	—	—	✓
Deployment	Hosted in US	Hosted in US	Add-on for self-hosted deployment in customer's VPC
Support			
Email Support	Community Discord	✓	✓
Shared Slack Channel Support	—	—	✓
Team Trainings	—	—	✓
Application Architectural Guidance	—	—	✓

LangSmith 平台功能介绍

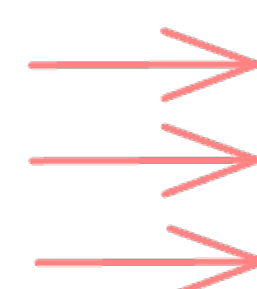
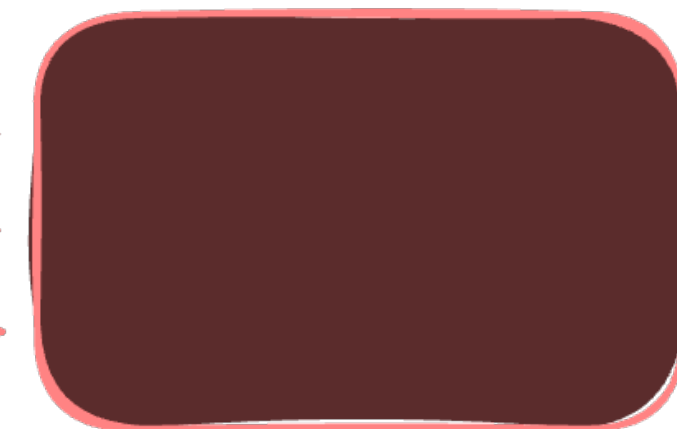
LangSmith Tracing

Project

Questions



RAG app



Answers

Collection of traces

Trace

Question



RAG app



Answer

Collection of runs

Run

Question



Indexing



Docs

Single unit of work

Docs



Generation



Answer

Tags, Feedback, Metadata



Retriever - LangSmith

Chat LangChain

smith.langchain.com/o/ebbf2eb-769b-4505-aca2-d11de10372a4/projects/p/f53ccf51-57c1-4c97-afdf-7ca5569945cd/r/2e7b107a-6686-47ea-93ae-fb753ceac369?trace_id=24d83889-c9...

LangChain Inc. > Projects > chat-langchain > /chat > ... > Retriever

PlaygroundAdd to DatasetShareAnnotate

Trace

/chat SUCCESS

3.63s 4,127 model:openai_gpt_3_5_turbo

Retriever 0.37s

ChatOpenAI 3.14s

Some runs have been hidden. [Show 15 hidden runs](#)

Retriever

RunFeedbackMetadata

Input

1 {

2 "query": "How do I use a RecursiveUrlLoader to load content from a page?"

3 }

JSON

Output

DOCUMENTS 6

Recursive URL | Langchain [Skip to main content](#__docusau... Recursive URL | Langchain +1

langchain_community.document_loaders.recursive_url_loader.RecursiveUrlLoader — La... +2

lazy_load() Lazy load web pages. load() Load web pages. load_and_split([text_splitter]) Load D... +2

URL | Langchain [Skip to main content](#__docusaurus_skipToConten... URL | Langchain +1

WebBaseLoader | Langchain [Skip to main content](#__docu... WebBaseLoader | Langchain +1

langchain_community.document_loaders.news.NewsURLLoader — LangChain 0.1.4 Lang... +2

START TIME

02/13/2024, 12:15:14 AM

END TIME

02/13/2024, 12:15:14 AM

TIME TO FIRST TOKEN

N/A

STATUS

Success

TOTAL TOKENS

0 tokens

LATENCY

0.37s

TYPE

Retriever

TAGS

Weaviate seq:step:2 model:openai_gpt_3_5_turbo OpenAIEmbeddings

Retriever - LangSmith

Chat LangChain

smith.langchain.com/o/ebbf2eb-769b-4505-aca2-d11de10372a4/projects/p/f53ccf51-57c1-4c97-afdf-7ca5569945cd/r/2e7b107a-6686-47ea-93ae-fb753ceac369?trace_id=24d83889-c9...

LangChain Inc. > Projects > chat-langchain > /chat > ... > Retriever

PlaygroundAdd to DatasetShareAnnotate

Trace

CollapseStatsMost relevant

/chat SUCCESS

3.63s4,127model:openai_gpt_3_5_turbo

Retriever 0.37s

ChatOpenAI 3.14s

Some runs have been hidden. [Show 15 hidden runs](#)

Retriever

RunFeedbackMetadata

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DOCUMENTS 6

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lazy_load() Lazy load web pages. load() Load web pages. load_and_split([text_splitter]) Load D... +2

URL | Langchain [Skip to main content](#__docusaurus_skipToConten... URL | Langchain +1

WebBaseLoader | Langchain [Skip to main content](#__docu... WebBaseLoader | Langchain +1

langchain_community.document_loaders.news.NewsURLLoader — LangChain 0.1.4 Lang... +2

START TIME

02/13/2024, 12:15:14 AM

END TIME

02/13/2024, 12:15:14 AM

TIME TO FIRST TOKEN

N/A

STATUS

Success

TOTAL TOKENS

0 tokens

LATENCY

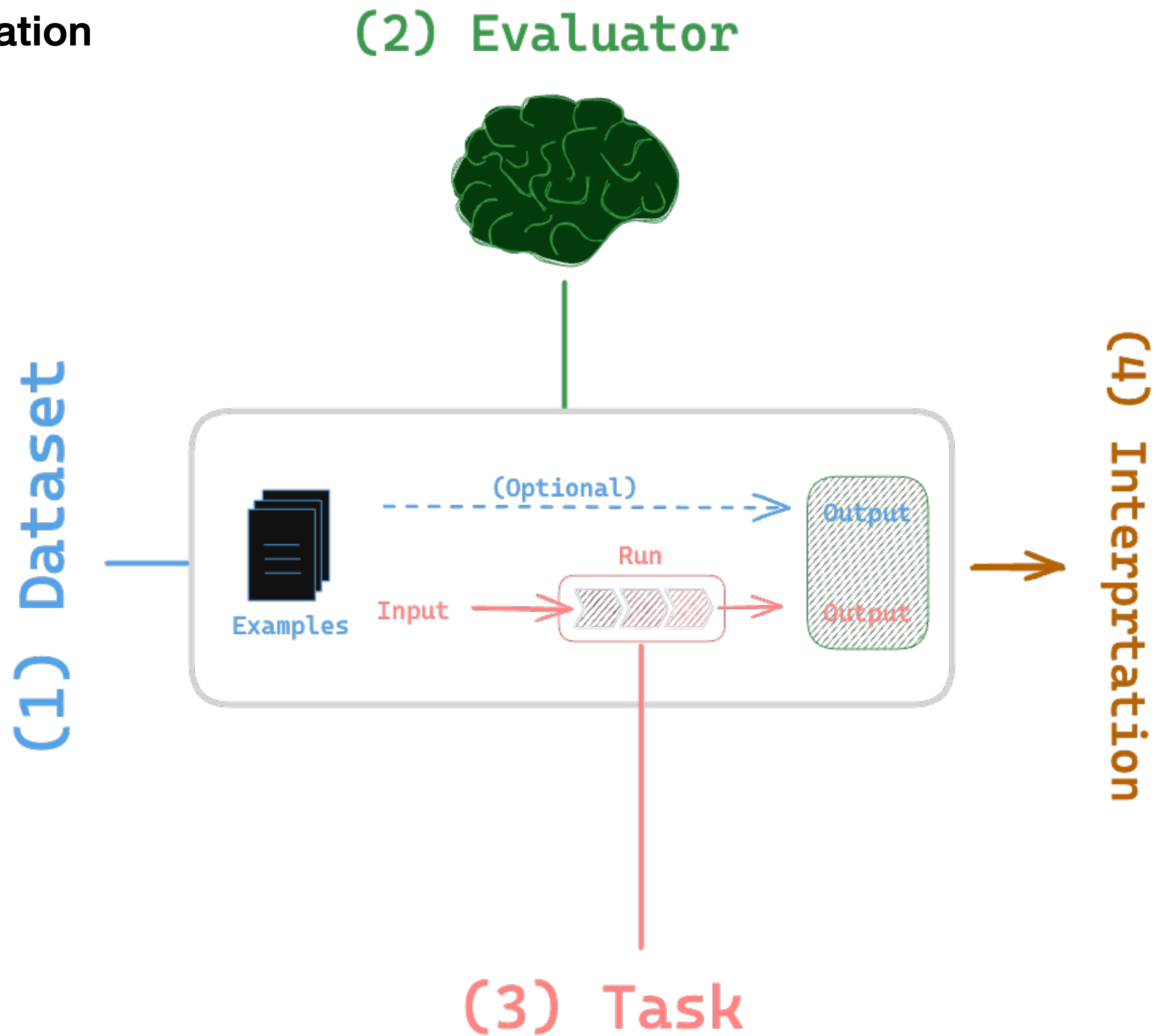
0.37s

TYPE

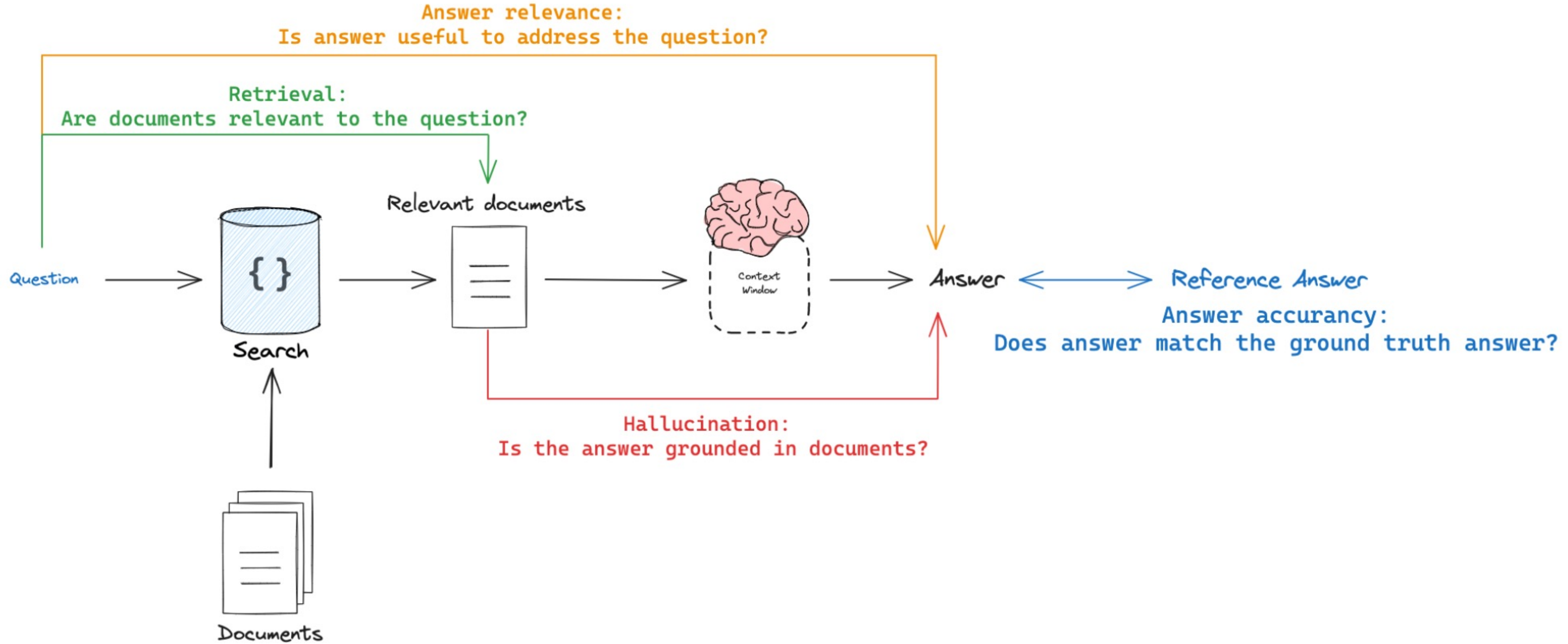
Retriever

TAGS

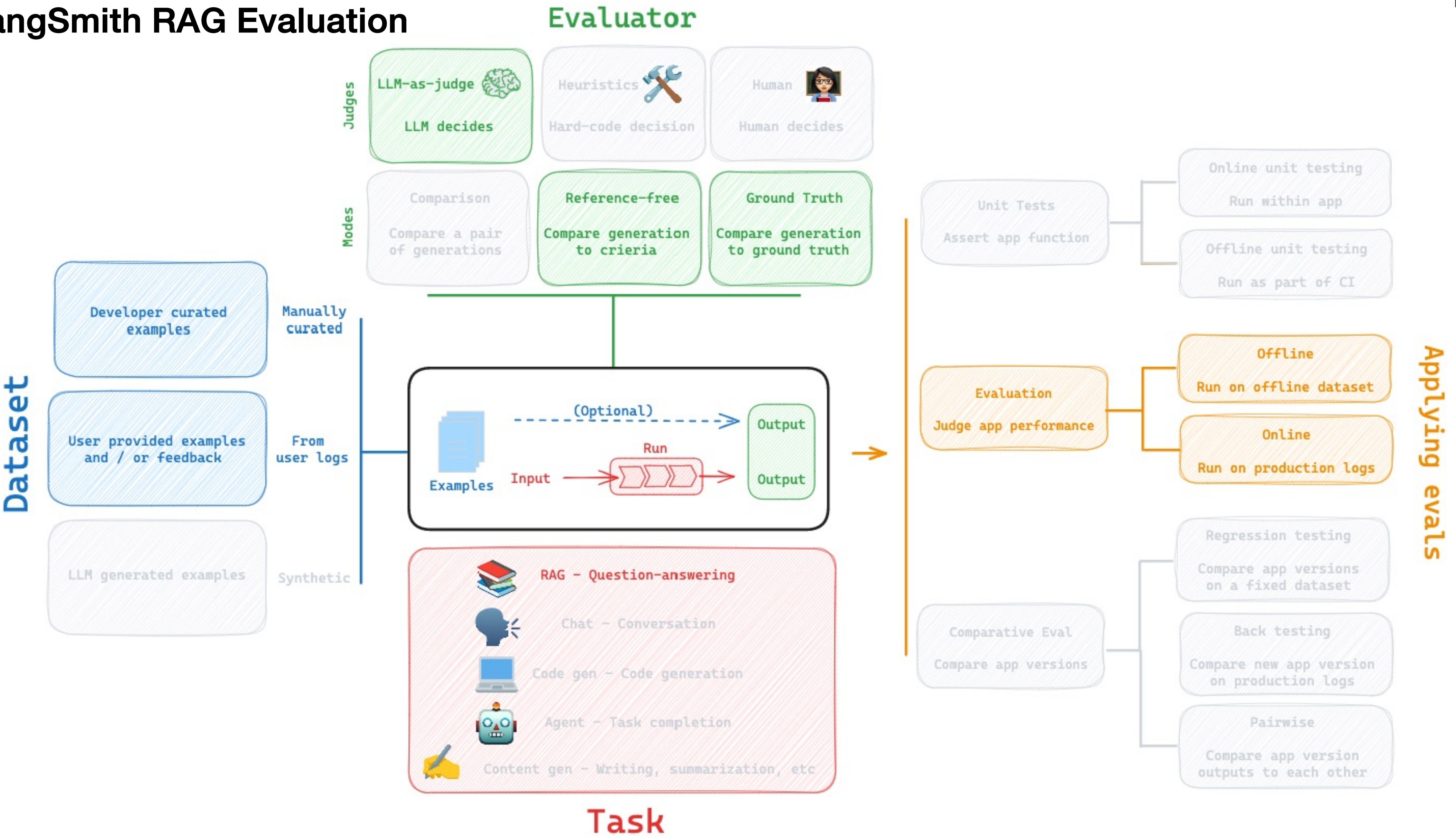
Weaviate seq:step:2 model:openai_gpt_3_5_turbo OpenAIEmbeddings



LangSmith RAG Evaluation



LangSmith RAG Evaluation



LangSmith 上手实战

DEMO

THANKS

 极客时间 | 训练营